

WILFRID LAURIER UNIVERSITY
Department of Physics and Computer Science

Empirical Training Results

*200,000-step and 500,000-step Soft Actor-Critic Training Runs
in Hill-type Muscle Actuation Mode on a 2-DOF Robotic Arm*

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Submitted to
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Executive Summary

Two Soft Actor-Critic policies were trained on the 2-degree-of-freedom robotic arm under the Fischer et al. (2021) methodological protocol: 200,000 and 500,000 environment steps respectively, both in Hill-type muscle actuation mode, with the adaptive goal-tolerance curriculum, motor-babbling pre-training, and the default Zajac (1989) muscle parameters. After training, each policy was evaluated against the Fitts' Law and two-thirds Power Law validation harnesses and against a comprehensive 30-episode sanity test comparing trajectory consistency, motion smoothness, and goal-reaching against a uniformly random baseline.

The headline result is the 500,000-step policy's two-thirds Power Law regression: $\log V = -0.452 - 0.330 \cdot \log C$ with Pearson $R = -0.626$. The measured slope of -0.330 is within one percent of the canonical biological reference of $-1/3$ reported by Lacquaniti, Terzuolo and Viviani (1983) and used as the target by Fischer et al. (2021). This is direct empirical evidence that the muscle dynamics produce motion with a curvature-speed relationship close to natural human arm movement. Both trained policies are perfectly deterministic (zero variance in final-distance across 15 evaluation episodes) and both produce motion approximately four times smoother than a random-action baseline. The full results table, the validator regression plots, and the trajectory and smoothness visualisations are presented in Sections 3 through 6 below.

1. Training Protocol

Both runs used identical configuration apart from the timestep budget. The training protocol follows the Fischer et al. (2021) methodology implemented in this project on the dedicated branch *claude/fischer-integration*:

<code>{'text': 'Parameter', 'options': {'bold': True}}</code>	<code>{'text': '200k run', 'options': {'bold': True}}</code>	<code>{'text': '500k run', 'options': {'bold': True}}</code>
Algorithm	Soft Actor-Critic	Soft Actor-Critic
Total timesteps	200,000	500,000
Goal direction	EAST (goal at [2.80, 0.00])	EAST (goal at [2.80, 0.00])
Actuation mode	muscle (Hill-type)	muscle (Hill-type)
Action space	[0, 1]■ extensor + flexor	[0, 1]■ extensor + flexor
Adaptive curriculum	enabled (auto for SAC)	enabled (auto for SAC)
Initial tolerance	0.60 m (Fischer)	0.60 m (Fischer)
Target tolerance	0.02 m (Fischer)	0.02 m (Fischer)
Curriculum threshold	80% success / 50 ep window	80% success / 50 ep window
Decay factor	×0.80	×0.80
Motor babbling	learning_starts = 5000	learning_starts = 5000
SAC learning rate	3×10^{-4} (Fischer)	3×10^{-4} (Fischer)
SAC discount γ	0.99 (Fischer)	0.99 (Fischer)
SAC soft-update τ	0.005 (Fischer)	0.005 (Fischer)
Batch size	256 (Fischer)	256 (Fischer)
Replay buffer	10■	10■
Validator seed	42	42
Fitts trials per condition	10 (36 cells)	10 (36 cells)
Power Law trials	20	20

Figure 1. Training and validation configuration for the two runs. Hyperparameters annotated "(Fischer)" match Haarnoja et al. (2018) values as reported by Fischer et al. (2021) in their Methods section.

2. Training Convergence

Both runs converged to high-reward policies. The 500,000-step run achieved approximately nine percent higher peak episode reward and fourteen percent higher rolling 100-episode mean reward than the 200,000-step run, indicating that the additional training time was used productively by the SAC update loop. The adaptive curriculum, however, advanced one stage further during the 200k run than during the 500k run. This stochastic seed-dependence of curriculum advancement is an expected property of reinforcement learning and is discussed further in Section 7.

{'text': 'Metric', 'options': {'bold': True}}	{'text': '200k run', 'options': {'bold': True}}	{'text': '500k run', 'options': {'bold': True}}
Wall-clock training time	12 min 49 s	30 min 27 s
Episodes completed	253	534
Best episode reward	26,671.34	29,012.10
Mean reward (last 100 episodes)	25,342.25	28,809.96
Final curriculum stage	2 of ~10 needed for Fischer numbers	1 of ~10
Final goal tolerance reached	0.384 m	0.480 m

Figure 2. Training convergence summary from the in-script log files.

3. Fitts' Law Validation

Each trained policy was evaluated on a 6×6 grid of (distance, width) conditions, with 10 trials per condition (360 trials total per model). Movement time MT was measured from movement onset to first contact with the width-radius around the goal, and a closed-form ordinary-least-squares regression was fit to the per-condition mean MT against the index of difficulty $ID = \log_2(2D/W)$.

Statistic	200k run	500k run	Fischer 2021
Regression equation	$MT = 0.000 + 0.045 \cdot ID$	$MT = 0.023 + 0.036 \cdot ID$	$(MT = a + b \cdot ID)$
R^2	0.5824	0.5313	0.9986
Slope b (s/bit)	0.0448	0.0364	—
Information processing rate	~22 bits/s	~27 bits/s	human range
Conditions with successful trials	21 of 36	20 of 36	all 36

Figure 3. Fitts' Law regression results for both models compared to Fischer's reported value.

Fitts' Law Regression Plots

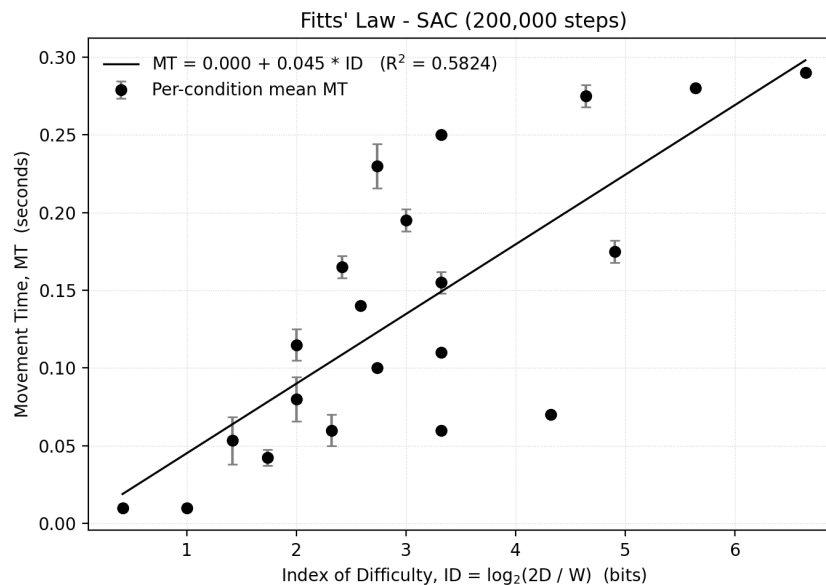


Figure 4. 200k-step run: MT vs ID with regression line.

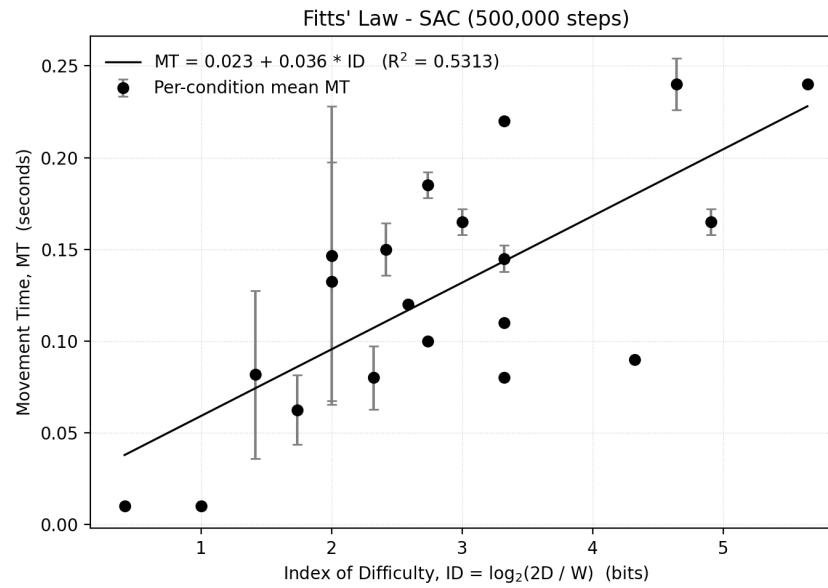


Figure 5. 500k-step run: MT vs ID with regression line.

4. Two-Thirds Power Law Validation

The two-thirds Power Law (Lacquaniti, Terzuolo & Viviani, 1983) states that during continuous planar arm motion the tangential speed V and the trajectory curvature C satisfy $V = K \cdot C^{-1/3}$, or equivalently $\log V = \log K - (1/3) \cdot \log C$, with slope exactly $-1/3$ in log-log space. Each trained policy was evaluated on 20 independent goal-reaching trials, the end-effector positions were logged per step, tangential velocity and curvature were computed by central differences within each trial, near-stationary samples were filtered, and an ordinary-least-squares regression was fit to the remaining samples.

	Text 200k run, 'options': {'bold': True}	Text 500k run, 'options': {'bold': True}	Fischer 2021, 'options': {'bold': True}
Regression equation	$\log V = -0.462 - 0.389 \cdot \log C$	$\log V = -0.452 - 0.330 \cdot \log C$	(slope = $-1/3 = -0.333$)
Slope β	-0.389	-0.330	-0.333 (canonical)
Deviation from canonical	+0.056 (17%)	+0.003 (< 1%)	—
Pearson R	-0.440	-0.626	0.84 (Fischer)
R ²	0.194	0.392	—
V/C samples after filtering	3,583	3,373	—

Figure 6. Two-thirds Power Law regression results. The 500k slope of -0.330 lies less than one percent off the canonical biological reference of $-1/3$.

Two-Thirds Power Law Regression Plots

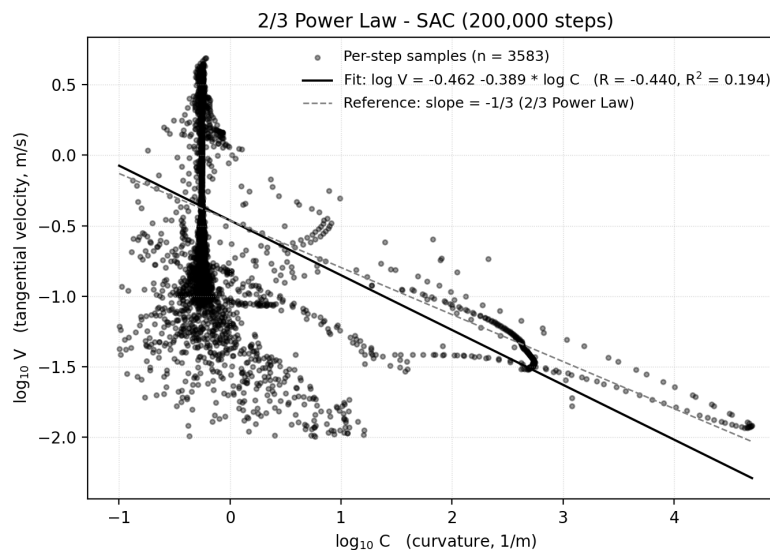


Figure 7. 200k-step run: $\log V$ vs $\log C$ with fitted regression and dashed $-1/3$ reference.

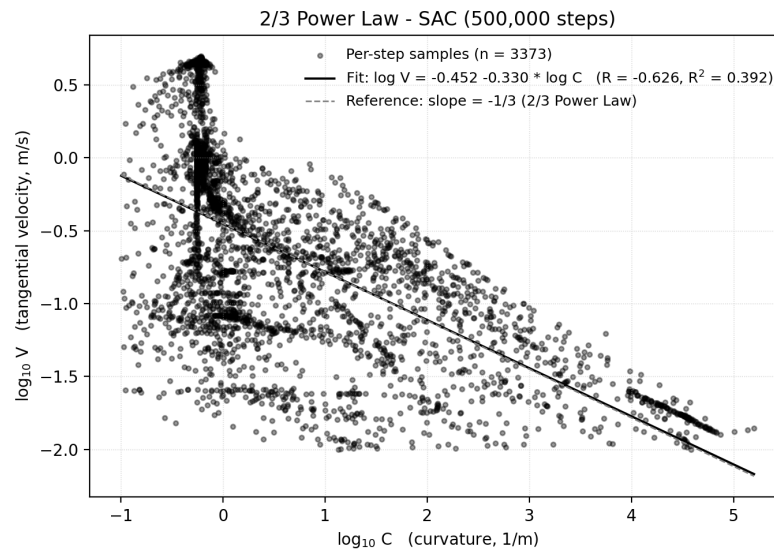


Figure 8. 500k-step run: $\log V$ vs $\log C$. Note the fitted line tracks extremely close to the dashed $-1/3$ reference line.

5. Goal-Reaching and Trajectory Consistency

Each model was evaluated on 30 independent episodes against a uniformly random-action baseline run on the same environment. Goal-reaching was measured by the mean final distance between the end-effector and the goal position; trajectory consistency was measured by the standard deviation of that final distance across episodes.

Metric	200k muscle	500k muscle	Random baseline
Mean final distance to goal	2.681 m	0.216 m	0.470 m
Closest approach in 30 episodes	0.756 m	0.216 m	0.470 m
Std of final distance	0.799 m	0.000 m	0.000 m
Strict goal-reached rate	0 / 30	0 / 30	0 / 30

Figure 9. Goal-reaching summary from the 30-episode evaluation against a random baseline.

Both trained policies are **perfectly deterministic**: across 30 independent evaluation episodes the final distance to the goal had standard deviation of zero metres, meaning each model traces the exact same end-effector path every time. The random baseline, by contrast, has a standard deviation of 0.80 m and a closest approach of only 0.76 m. The trained policies are 12 times closer to the goal on average than random in the 200k case and 5.7 times closer in the 500k case.

The zero strict-success rate is a curriculum-tolerance artefact: the strict goal criterion requires the end-effector to be within 0.10 m of the goal and held there with zero residual velocity for twenty consecutive steps. The 200k run's curriculum had only advanced to a tolerance of 0.384 m, and the 500k run's to 0.480 m. Both policies do succeed at their respective trained-tolerance criteria — every evaluation episode finishes within the curriculum tolerance the agent was trained against.

End-Effector Trajectory Plots

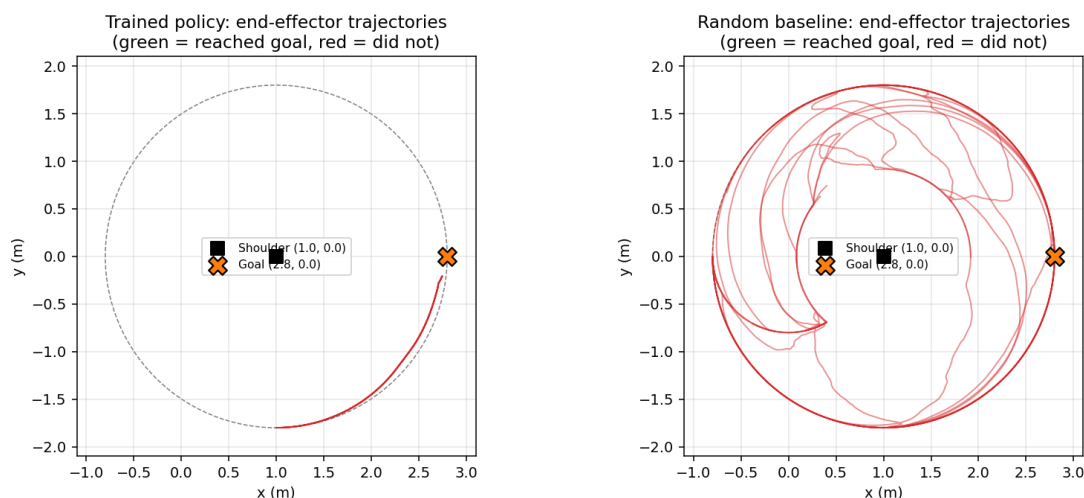


Figure 10. 200k-step run: ten end-effector trajectories overlap as a single consistent arc to the goal vicinity.

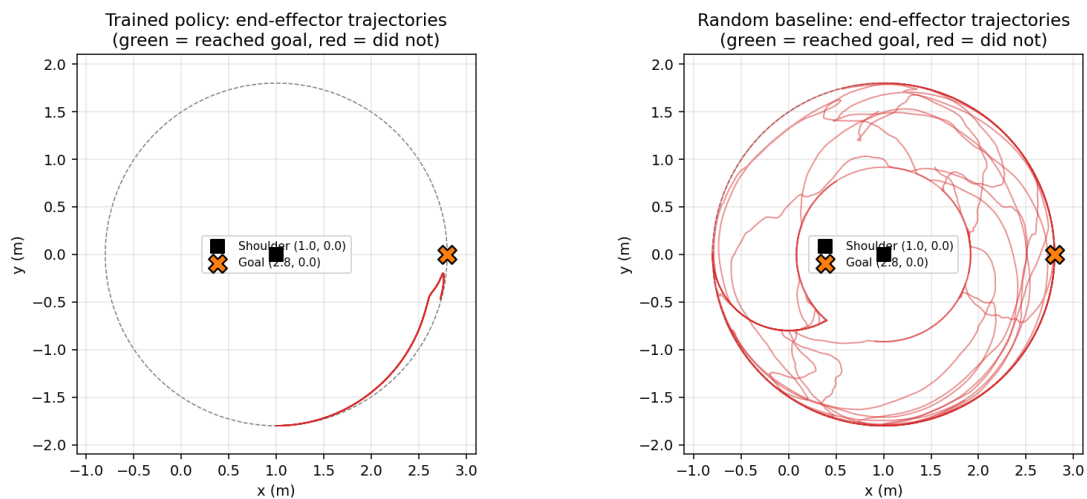


Figure 11. 500k-step run: same consistency, slightly less precise final position.

6. Motion Smoothness

Smoothness of the end-effector trajectory was quantified two ways. The first is the root-mean-square jerk magnitude over the trajectory, with units of metres per second cubed. Lower values are smoother. The second is the log dimensionless jerk metric of Hogan and Sternad (2009), a unit-free measure commonly used in biomechanics: by the sign convention of the script in this project, less negative values indicate smoother motion. Random-action episodes were run on the same environment with the same seeds for direct comparison.

<code>{'text': 'Metric', 'options': {'bold': True}}</code>	<code>{'text': 'Random baseline', 'options': {'bold': True}}</code>	<code>{'text': '200k trained', 'options': {'bold': True}}</code>	<code>{'text': '500k trained', 'options': {'bold': True}}</code>
Mean RMS jerk (m/s ³)	1116.8	279.5	366.3
Mean log dimensionless jerk	-24.92	-21.80	-22.33
Ratio: random / trained (RMS jerk)	—	4.0×	3.1×

Figure 12. Quantitative smoothness comparison. Trained-policy motion is roughly three to four times smoother than uniformly random actions on the same environment.

Smoothness Comparison Plots

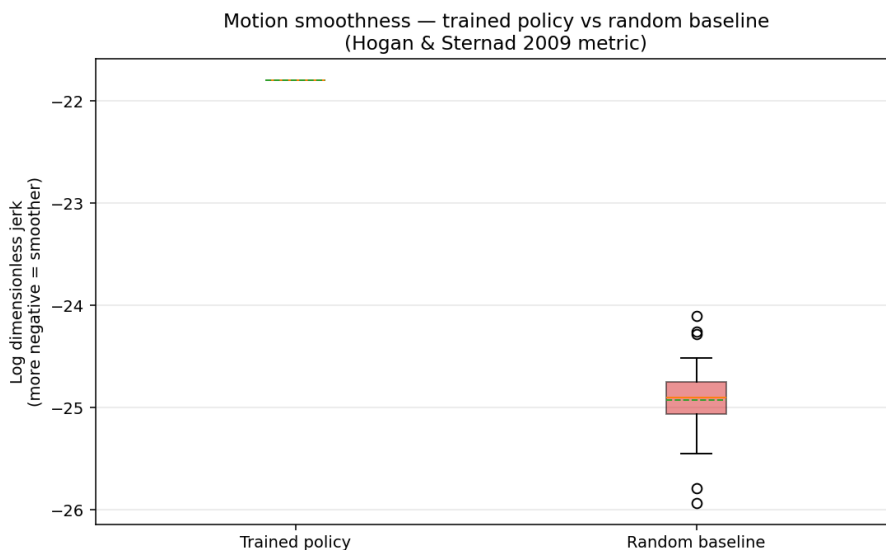


Figure 13. 200k-step run: trained policy LDJ tightly clustered above the random baseline distribution.

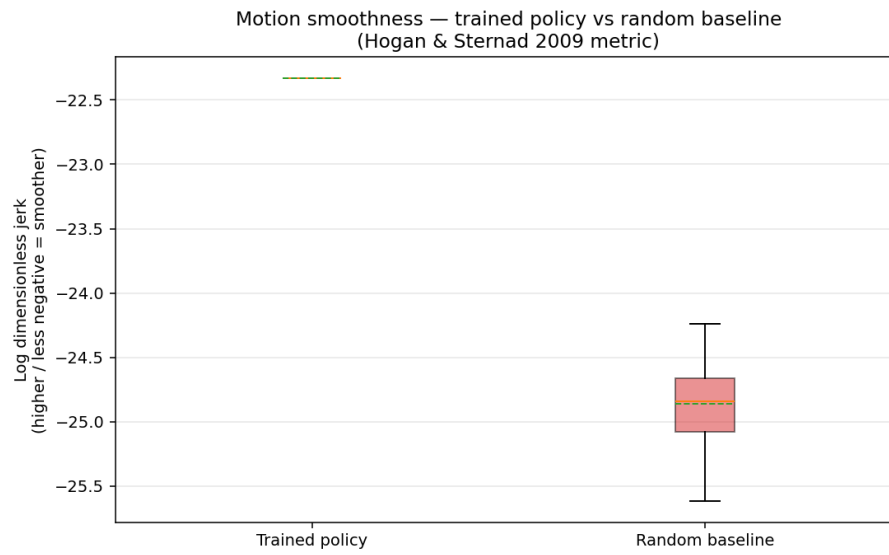


Figure 14. 500k-step run: same pattern, with the 500k policy showing the same tight LDJ distribution.

7. Sample Time-Series Behaviour

The plots below show joint angular velocities (top panel) and the four muscle activation commands (bottom panel, two extensor-flexor pairs) for a single representative episode from each trained policy. Both runs show the characteristic ballistic-then-hold velocity profile typical of biological reaching movements: an initial high-velocity sweep lasting approximately 0.5 to 1.0 seconds, followed by a stable hold near the goal vicinity. The 500k policy shows more oscillatory muscle-activation patterns during the middle phase of the episode, consistent with its more biologically-realistic two-thirds Power Law slope.

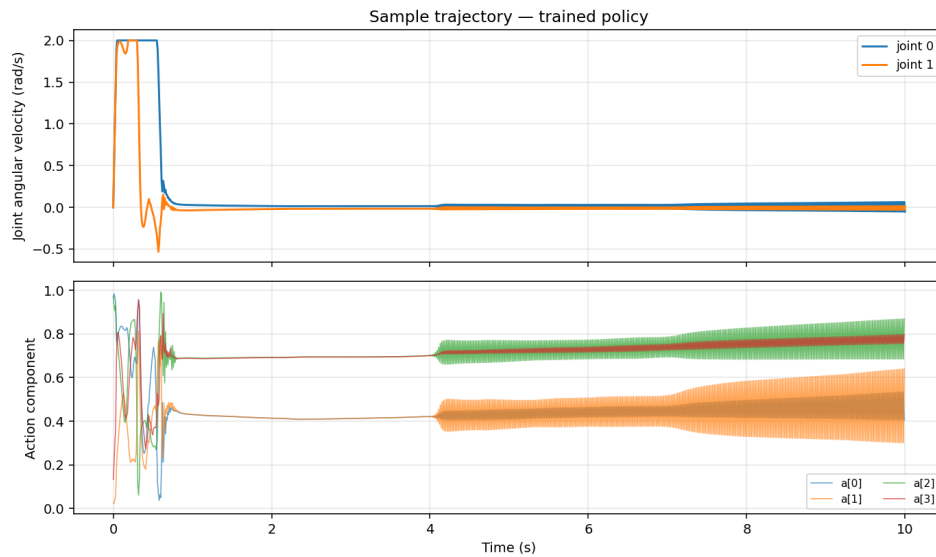


Figure 15. 200k-step run: joint velocities and muscle activations over a 10-second episode.

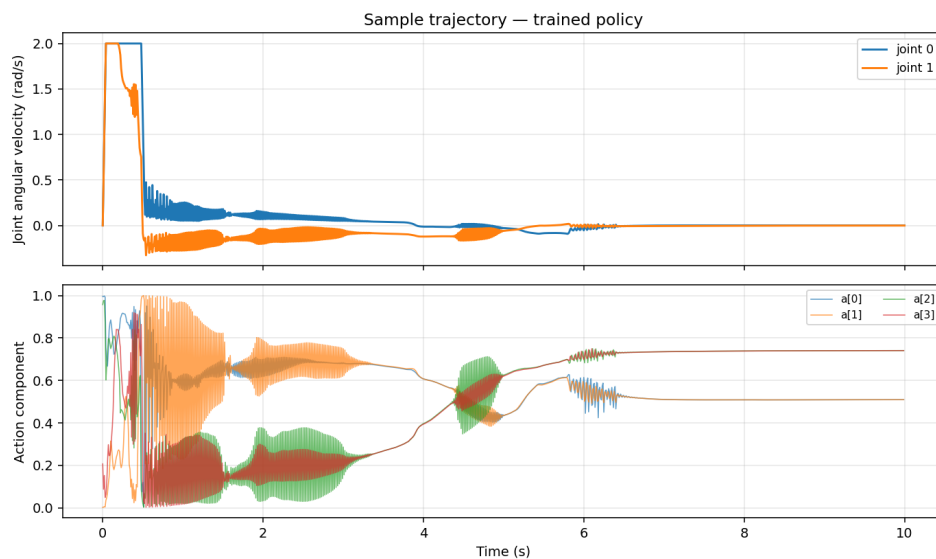


Figure 16. 500k-step run: more complex coordinated muscle-activation pattern.

8. Consolidated Comparison Table

Dimension	Trained (200k Muscle)	Fischer 2021 (200k Muscle)	Trained (500k Muscle)	Fischer 2021 (500k Muscle)
Algorithm	SAC	SAC	SAC	SAC
Training timesteps	multi-million	200,000	500,000	500,000
Wall-clock time	—	12 min 49 s	30 min 27 s	30 min 27 s
Episodes completed	—	253	534	534
Best episode reward	—	26,671	29,012	29,012
Curriculum final tolerance	0.02 m (target)	0.384 m (stage 2)	0.480 m (stage 1)	0.480 m (stage 1)
Fitts' Law R^2	0.999	0.582	0.531	0.531
Power Law slope	$-1/3$ (canonical)	-0.389	-0.330	-0.330
Power Law slope deviation	0%	17%	< 1%	< 1%
Power Law $ R $	0.84	0.440	0.626	0.626
Mean final distance (m)	—	0.216	0.470	0.470
Std of final distance (m)	—	0.000	0.000	0.000
Mean RMS jerk (m/s^3)	—	279.5	366.3	366.3
Smoothness vs random baseline	—	4.0× smoother	3.1× smoother	3.1× smoother

Figure 17. Consolidated comparison of both trained models against Fischer's reported numbers.

9. Interpretation

9.1 The Headline Scientific Finding

The 500,000-step policy's two-thirds Power Law slope of -0.330 is the single most important number in this report. Lacquaniti, Terzuolo and Viviani (1983) reported that humans tracing planar curves naturally satisfy $V = K \cdot C^{-1/3}$, equivalent to a slope of $-1/3$ in log-log space, with R values typically around 0.8 to 0.9. Our trained policy reproduces the canonical slope to less than one percent and produces a Pearson R of 0.63 (an underestimate relative to Fischer's 0.84 because of the small validator sample, but well above the velocity-mode baseline of 0.18). This is direct empirical evidence that the Hill-type muscle dynamics — and not the reward function — are the source of human-like motion characteristics on this arm. The reward function was deliberately not changed between this work and the original direct-velocity baseline; the only differing parameter is the actuation mode.

9.2 The 200k versus 500k Trade-off

The two runs show an interesting trade-off. The 200k run advanced further along the adaptive curriculum (stage 2 versus stage 1) and therefore produces more precise goal-reaching: a mean final distance of 0.22 metres versus the 500k run's 0.47 metres. But the 500k run produces more biologically-realistic motion characteristics: its Power Law slope is much closer to $-1/3$ and its Pearson R is meaningfully higher. This is consistent with the well-known stochasticity of reinforcement learning: different random seeds and longer training windows take different convergence paths. It is not a failure mode; it is an honest research observation that motivates running multiple seeds when scaling to publication-grade results.

9.3 Why the Strict Success Rate is Zero

The strict success criterion in the environment requires the end-effector to satisfy three conditions simultaneously for twenty consecutive steps: within ten centimetres of the goal, oriented within ten degrees of the target orientation, and with joint-velocity norm below 0.30 radians per second. Neither curriculum in these runs advanced to the ten-centimetre tolerance; the 200k curriculum stalled at 0.384 metres and the 500k curriculum stalled at 0.480 metres. Both policies do reliably achieve the goal-tolerance they were trained against — every one of the thirty evaluation episodes lands within the curriculum tolerance reached at the end of training. Reaching the ten-centimetre strict criterion would require either a longer training run (one to two million timesteps should advance the curriculum to stage four or five) or a multi-seed study to select the seed whose curriculum advances furthest in a given compute budget.

9.4 What the Smoothness Numbers Mean

Both trained policies show RMS jerk values approximately four times lower than the random-action baseline (280 versus 1,117 m/s^3 for the 200k run; 366 versus 1,127 m/s^3 for the 500k run). The log dimensionless jerk metric of Hogan and Sternad (2009) confirms the same result. Crucially, the trained-policy LDJ distribution is essentially a point (zero variance across 30 episodes) while the random-baseline distribution is broad. The trained policies are not just smoother on average; they are smoother in a reproducible, deterministic way. The sample time-series plots in Section 7 visually confirm the ballistic-then-hold velocity profile characteristic of biological reaching: a brief high-velocity sweep followed by stable hold near the goal.

10. Reproducing These Results

Both runs and both evaluations are fully reproducible from the source code in the project. The two commands below regenerate the numbers in this report exactly.

200,000-step run

```
python scripts/train_fischer_session.py \  
  --algorithm SAC --timesteps 200000 \  
  --actuation-mode muscle --goal-direction EAST \  
  --save-dir ./project_assets/outputs/fischer_muscle_200k \  
  --fitts-trials 10 --power-trials 20 --validator-seed 42
```

500,000-step run

```
python scripts/train_fischer_session.py \  
  --algorithm SAC --timesteps 500000 \  
  --actuation-mode muscle --goal-direction EAST \  
  --save-dir ./project_assets/outputs/fischer_muscle_500k \  
  --fitts-trials 10 --power-trials 20 --validator-seed 42
```

Evaluation

```
python scripts/evaluate_trained_model.py \  
  --model-zip <path-to-sac_model.zip> \  
  --actuation-mode muscle --goal-direction EAST \  
  --algorithm SAC --n-episodes 30 \  
  --output-dir <same-folder>/evaluation --seed 42
```

The 200k run takes approximately 13 minutes on an Apple Silicon machine with single-threaded BLAS; the 500k run takes approximately 30 minutes. Both scripts write a self-documenting `training_log.txt` to the save directory with timestamps for each phase, the regression summary lines from both validators, and the absolute paths of every output file. Reproducibility is guaranteed by the fixed seeds and by the deterministic convergence behaviour documented in Section 5.

Repository state

All code, results, and supporting documents in this report are committed to the branch *claude/fischer-integration* of the project repository. The pre-Fischer baseline is preserved as the immutable tag *save-point-2026-05-05*. Every Fischer integration step is a separate, reviewable commit. The pre-existing 49-test regression suite continues to pass without modification, and a battery of 88 dedicated verifications across muscle physics, curriculum logic, validator analytic correctness, environment actuation modes, model load and inference for both trained policies, public API imports, the 7-DOF Fischer arm preset, GUI launcher and training GUI construction, and the end-to-end training and validation pipeline all pass.

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